juddles

Track • Plan • Grow

Product requirements, UI/UX system and technical implementation blueprint



Visual reference: five-screen Juddles concept supplied for this specification. Use it as design direction, not as a pixel-for-pixel template.

A single place to understand personal progress and money: goals, income, expenses, budgets, cash flow and financial performance.

Document purpose Define the product scope and provide a practical implementation plan for a production-ready mobile app. This document describes requirements; it does not claim that the app or backend has already been built.

Design direction Calm, human, minimal and tactile. Warm cream surfaces, cool mint accents, deep evergreen anchors, pill-shaped controls, circular indicators and a compact floating navigation bar.

1. Product overview

Juddles is a personal progress and money-management app. It combines everyday financial tracking with goal planning so users can understand what came in, what went out, what remains, and whether they are moving toward their goals.

Product principles

• **Clarity before complexity:** show the most useful financial picture first; keep advanced breakdowns one tap away.

• **Fast capture:** adding an income or expense should take only a few deliberate inputs.

• **Real data only:** all totals, charts and progress indicators must derive from saved user records. Never populate production screens with demo transactions.

• **User control:** let users choose their currency, budget periods, categories, privacy settings and notification preferences.

• **Honest calculations:** clearly distinguish cash balance, net cash flow, savings and profit-like surplus. Do not label a number as profit unless its definition is explicit.

Primary audience and use cases

User need	Juddles response
Know where money stands	A home dashboard with current balance, income, expenses and net cash flow.
Control spending	Monthly and custom-period budgets with category-level progress and alerts.
Save toward something	Goals with target amounts, deadlines, contributions and completion status.
Review performance	Reports for trends, category breakdowns, income vs. expenses and savings rate.
Track life progress	Non-financial goals and milestones can be included alongside financial goals.

Suggested platform

Mobile-first Android and iOS experience, with Firebase as the backend foundation. A responsive web dashboard can be added later without changing the underlying data model.

2. Visual identity and UI/UX system

The supplied reference establishes the visual language. The improved concept should feel like a thoughtfully art-directed finance product rather than a generic dashboard template.

Color system

Token	Suggested value	Usage
Evergreen	#075B43	Primary actions, high-priority cards, selected navigation
Fresh green	#16845F	Progress, positive movement, charts
Mint	#DDF3E4	Soft surfaces, progress backgrounds, selected states
Ice green	#EEF7F2	Secondary surfaces and subtle section separation
Warm cream	#F7F5EA	Main background; reduces stark white glare
Ink	#17231F	Primary text and high-contrast labels
Muted	#64736B	Supporting labels, dates and helper text
Coral	#E98A68	Expense cues and caution states; use sparingly

Shape, spacing and components

• **Cards:** 20–26 dp corner radius, soft low-elevation shadow, consistent inner padding. Avoid stacking too many cards in one viewport.

• **Pills:** use for date filters, category chips, segmented tabs and primary actions. Keep labels short and tap targets at least 44–48 dp high.

• **Floating navigation:** a rounded capsule above the bottom safe area. Use four destinations plus one prominent central add button where appropriate.

• **Progress:** use circular rings for one headline metric and slim horizontal bars for lists. Always pair progress visuals with readable values.

• **Charts:** minimal gridlines, clear legends, accessible labels and selectable periods. Do not rely on color alone.

• **Motion:** subtle transitions, count-up only when a value changes, gentle progress animation and clear pressed/loading states. Respect reduced-motion settings.

• **Typography:** use a highly legible system sans-serif. Establish a compact hierarchy: screen title, key number, section title, supporting text.

Layout rules

• Adapt to small, standard and large phones using responsive constraints and safe-area insets; do not hard-code one screen size.

• Prevent overlap by using scrollable content, intrinsic sizing and tested minimum widths. Long labels must wrap or truncate intentionally.

• Use a compact mode setting to reduce vertical spacing and card height without hiding essential information.

• Keep the main action reachable with one hand; floating elements must not cover content or system navigation.

• Use skeleton loading only while fetching real data, and provide explicit empty, offline, error and permission-denied states.

3. Five-screen experience

The five screens below map directly to the supplied mockup. The first screen is the sign-in entry point; the remaining screens form the main signed-in experience.

Screen	Purpose and content	Key interactions
1. Welcome / Login	Juddles identity, concise value statement, mint/cream illustration or abstract growth motif, Continue with Google, Continue with Email.	Google OAuth; email sign-up/sign-in; password reset; terms and privacy links; loading and error states.
2. Home dashboard	Total balance, income, expenses, net cash flow, monthly progress, quick actions, recent activity and goal summary.	Add income/expense; set budget; create goal; open reports; change date range; hide sensitive amounts.
3. Goals	Goal cards with icon, title, target, current amount, deadline, progress and milestone status.	Create/edit/archive goal; add contribution; change target/date; mark milestone; view history.
4. Expenses & budgets	Period selector, spending donut, category totals, budget usage, recent transactions.	Add/edit/delete transaction; set category budget; recurring expense; filter/search; attach note or receipt if enabled.
5. Reports	Balance trend, income vs. expenses, category comparisons, net cash flow and savings progress.	Choose period; compare months; inspect chart values; export report; share a user-approved summary.

Screen 1 — Welcome and authentication

Keep this screen emotionally inviting and visually quiet. Place the Juddles mark and a short benefit statement in the upper-middle area, followed by two clear sign-in choices. Google sign-in should be a branded button; email should open a focused form. Avoid a long onboarding questionnaire before the user can enter the app.

Screen 2 — Home dashboard

The dashboard should answer four questions quickly: what is my current balance, how much came in, how much went out, and how am I progressing? Use one dominant balance card, a small set of quick actions, one headline progress ring and a short recent-activity list. Allow users to reorder or hide optional widgets later.

Screen 3 — Goals

Support both money goals (emergency fund, travel, device, home) and general progress goals if the user enables them. Each goal has a target, current amount, optional due date, contribution history and status. Contributions must be recorded as transactions or linked transfers to avoid double counting.

Screen 4 — Expenses and budgets

Make category spending understandable at a glance. A donut chart is useful only when accompanied by category names and exact values. Budget status should show amount spent, limit, remaining amount and period. Warn before a budget is exceeded, but let users customize alert thresholds.

Screen 5 — Reports

Reports should show trends rather than decorative graphs. Every chart needs a selected period, clear units, accessible data labels and a way to inspect underlying transactions. Net cash flow should be calculated consistently as income minus expenses for the selected period.

4. Functional requirements

The product should be built around dependable record keeping. Every action that changes a financial number must create, update or reverse a stored record and refresh dependent summaries.

Transactions and cash flow

• Create income and expense records with amount, currency, date/time, category, account or wallet, optional note and optional merchant/payee.

• Support edit, delete with confirmation, and transaction history. For auditability, consider soft-delete or reversal records rather than silently erasing posted entries.

• Support recurring records with frequency, start date, optional end date and next occurrence. Make it clear whether a recurring item is scheduled or already posted.

• Allow transfers between user-owned accounts. Transfers must not be counted as income or expenses in consolidated reports.

• Provide search and filters by date, category, account, transaction type and amount range.

• Handle decimal precision and currency-specific minor units. Store amounts in integer minor units where practical.

Budgets

• Create overall and category budgets for weekly, monthly, yearly or custom periods.

• Track spent, remaining and percentage used from eligible expense transactions.

• Allow rollover as an explicit setting; default should be off unless the user chooses otherwise.

• Support configurable notifications at thresholds such as 70%, 90% and 100%, with user-controlled settings.

• Prevent duplicate counting when transactions are edited, refunded or moved between categories.

Goals and progress tracking

• Create a goal with title, target amount or milestone count, current progress, target date, icon/color and optional notes.

• Record contributions and milestone events with timestamps; display a timeline of progress.

• Support pause, archive and completion states. Editing a target should recalculate percentage without rewriting past contribution history.

• Offer optional reminders based on a user-selected schedule; never send reminders without permission.

Income, surplus and profit terminology

For a personal finance tracker, prefer net cash flow or surplus over the word profit. Define net cash flow as total income minus total expenses for the chosen period. If the app later supports business accounts, add a separate profit calculation with explicit revenue, cost of goods, operating expenses, tax and adjustment rules.

5. Data model and calculation rules

Use a clear owner-scoped data model. Every financial record must belong to an authenticated user and include timestamps, currency and a stable identifier.

Collection / entity	Core fields
users	uid, displayName, email, photoURL, preferredCurrency, locale, timezone, createdAt, updatedAt
accounts	accountId, uid, name, type, currency, openingBalance, isArchived
transactions	transactionId, uid, accountId, type, amountMinor, currency, categoryId, occurredAt, note, merchant, recurrenceId, status, createdAt, updatedAt
categories	categoryId, uid or system scope, name, type, iconKey, colorToken, isArchived
budgets	budgetId, uid, name, periodType, startDate, endDate, currency, limitMinor, categoryIds, rolloverEnabled
goals	goalId, uid, title, targetMinor, currentMinor, currency, targetDate, status, iconKey, createdAt
goalEvents	eventId, uid, goalId, eventType, amountMinor, occurredAt, note, linkedTransactionId
recurringRules	ruleId, uid, transactionTemplate, frequency, startDate, endDate, nextRunAt, active
auditEvents	eventId, uid, entityType, entityId, action, timestamp, metadata

Calculation definitions

Metric	Definition
Income	Sum of posted income transactions in the selected period and currency.
Expenses	Sum of posted expense transactions in the selected period and currency.
Net cash flow	Income – expenses for the selected period.
Current account balance	Opening balance + posted income – posted expenses + incoming transfers – outgoing transfers, per account.
Budget remaining	Budget limit – eligible posted expenses in the budget period.
Budget usage	Eligible posted expenses ÷ budget limit × 100. Handle zero limits explicitly.
Goal progress	Current goal amount ÷ target amount × 100, capped visually at 100% while preserving overfunding in the value.
Savings rate	(Income – expenses) ÷ income × 100; show as unavailable when income is zero.

Currency handling

Allow the user to choose a primary display currency. Each transaction retains its original currency. If multiple currencies are supported in a single summary, store the exchange-rate source and rate timestamp used for conversion. Never add unlike currencies without conversion. For an initial release, a single-currency workspace is simpler and safer.

6. Technical architecture (Firebase)

A Firebase-backed architecture is a practical starting point for authentication, synchronized records and push notifications. Keep business logic and access rules explicit; do not rely on hiding controls in the UI for security.

Recommended stack

Layer	Recommended approach
Mobile UI	Native Android with Kotlin + Jetpack Compose, or a cross-platform client if iOS is required at launch. Keep UI state separate from data and domain logic.
Authentication	Firebase Authentication: Google sign-in and email/password. Add email verification, password reset and account deletion.
Database	Cloud Firestore with user-scoped documents, indexes for date/category queries and server timestamps.
Backend logic	Cloud Functions for recurring transaction generation, scheduled summaries, exports and trusted calculations where needed.
Files	Firebase Storage for optional receipt images or exported documents, with strict owner-scoped rules.
Notifications	Firebase Cloud Messaging for opt-in budget alerts, goal reminders and recurring-item notices.
Analytics / crash	Firebase Analytics and Crashlytics only with appropriate disclosure and privacy configuration.

Security and privacy requirements

• Firestore and Storage security rules must enforce request.auth.uid ownership for every private record.

• Validate transaction type, amount, currency and ownership on trusted server paths where appropriate.

• Never place service-account credentials or privileged secrets in the mobile app.

• Use App Check where supported; rate-limit sensitive or costly callable operations.

• Support secure sign-out, account deletion, data export and clear privacy disclosures.

• Do not log transaction notes, exact amounts or other sensitive financial details to analytics or crash logs.

• Provide local device authentication as an optional privacy layer, not as a substitute for server authorization.

• Use encrypted transport and rely on platform-managed secure storage for tokens and local secrets.

Offline and synchronization behavior

Define which operations are available offline. If users can add transactions offline, assign stable client-generated IDs, queue writes safely and resolve conflicts deterministically. Display pending-sync status. Do not present an unsynchronized value as confirmed server state without a visible indicator.

7. Core user flows and states

First-time user

• Open Juddles → welcome screen → choose Google or email.

• Authenticate → verify email if required → choose primary currency and timezone.

• Optional short setup: add opening balance, choose starter categories and create a first goal. Every setup step must be skippable.

• Land on an honest empty dashboard with a clear first action such as Add income or Add expense.

Add expense

• Tap the central Add action → choose Expense → enter amount and currency.

• Select category and account; date defaults to now but can be changed.

• Optional note or merchant → review summary → save.

• Update transaction list, category total, budget progress, account balance and reports from the saved record.

Create a budget

• Choose overall or category budget → enter limit, currency and period.

• Choose optional alert thresholds and rollover preference → save.

• Show current spend and remaining amount based on transactions in the budget period.

Required states on every relevant screen

State	Expected behavior
Loading	Skeleton or restrained progress indicator; prevent duplicate submissions.
Empty	Explain what belongs here and provide one clear next action.
Error	Plain-language message, retry path and no silent data loss.
Offline	Show last sync time and pending changes; avoid implying data is current.
Permission denied	Explain why permission is useful and offer a settings route; app remains usable where possible.
Success	Confirm saved action and refresh dependent totals.
Destructive action	Ask for confirmation and explain whether the action can be undone.

8. Settings, accessibility and quality

Settings and personalization

• Profile and account management, Google/email account details, sign-out and account deletion.

• Primary currency, number/date format, timezone and language.

• Theme: light, dark and system; preserve the mint/cream Juddles identity in light mode and create a carefully tuned dark palette.

• Compact mode: reduce spacing and card height while preserving legibility and touch targets.

• Notification controls for budget thresholds, goal reminders and recurring transactions.

• Data export (CSV and PDF initially), privacy information, support and app version.

Accessibility

• Maintain readable contrast for text and chart labels; test mint text against cream backgrounds.

• Support dynamic font scaling without clipping or overlapping components.

• Provide semantic labels for icons, charts, buttons and progress indicators.

• Use touch targets of at least 44–48 dp and avoid color-only status communication.

• Respect screen readers, reduced motion and device accessibility settings.

Quality and acceptance criteria

• All financial totals reconcile with the underlying transaction records for tested date ranges.

• Editing or deleting a transaction updates every affected dashboard, budget, goal link and report.

• Transfers do not inflate income or expenses in consolidated summaries.

• Currency mismatches are blocked or converted using a recorded exchange rate.

• Authentication and Firestore rules prevent one user from reading or modifying another user's data.

• The five key screens render without overlap on narrow, standard and large phone sizes and with large system text.

• Every button has a real action, loading state, error state and success response; no decorative non-functional controls.

• No seeded sample transactions or fake chart values appear in production. Any optional sample workspace must be clearly labelled and removable.

9. Delivery roadmap

Phase	Scope	Exit criteria
Phase 1 — Foundation	Design system, navigation, authentication, user profile, currency/timezone setup, Firestore rules.	User can securely sign in, configure preferences and reach a real empty dashboard.
Phase 2 — Money tracking	Accounts, income/expense transactions, categories, transfers, search and transaction history.	Balances and cash flow reconcile against saved records.
Phase 3 — Budgets and goals	Budget periods, category limits, goal creation, contributions, milestones and reminders.	Budget and goal progress update correctly after every relevant transaction.
Phase 4 — Insights	Reports, trend charts, comparisons, savings rate, export and filters.	Reports match the transaction ledger and clearly state period/currency.
Phase 5 — Hardening	Offline handling, security review, accessibility, performance, device-size QA and release preparation.	Critical tests pass; privacy, deletion and export flows are verified.

Suggested release boundary

For a dependable first release, prioritize authentication, one primary currency per workspace, manual income/expense entry, account balance, category budgets, financial goals and basic monthly reports. Add receipt scanning, bank integrations, multi-currency conversion and advanced forecasting only after the core ledger is reliable.

Definition of done

• A feature is complete only when its UI, persistence, validation, security rules, loading/error/empty states and tests are implemented.

• All financial calculations are covered by unit tests, including refunds, transfers, date boundaries, zero values and rounding.

• A release candidate is tested on multiple Android screen sizes and at least one low-memory device profile.

• The app includes clear privacy, data export and account deletion paths before public release.

10. Final design and product direction

Juddles should feel like a calm daily companion for progress and money management. The visual language should remain minimal and warm: cream backgrounds, mint surfaces, evergreen emphasis, rounded cards, pill controls and a floating navigation capsule. The product's credibility comes from accurate numbers, transparent calculations and reliable data handling—not from decorative charts or simulated activity.

Use the included five-screen mockup as the visual north star. Preserve its overall tone and interaction patterns while refining each screen for real content, accessibility, responsiveness and production behavior.

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